Experiment No. 08.

Aim: Applied Optimization.

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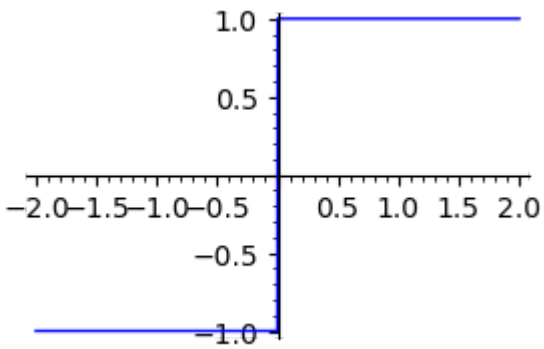
In [123]:
```
f(x)=x/abs(x)
plot(f,(x,-2,2),figsize=3)
```

Out[123]:



In [124]:
```
limit(f,x=0)
```

Out[124]: x |--> und

In [125]:
```
limit(f,x=0,dir="+")
```

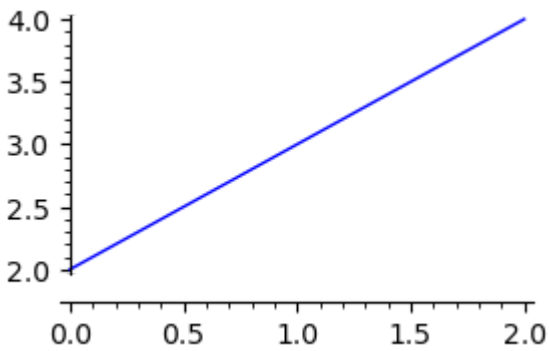
Out[125]: x |--> 1

In [126]:
```
limit(f,x=0,dir="-")
```

Out[126]: x |--> -1

In [127]:
```
f(x)=(x^2-4)/(x-2)
plot(f,(x,0,2),figsize=3)
```

Out[127]:



In [128]:
```
limit(f,x=2,dir="+")
```
Out[128]:
```
x |--> 4
```

In [129]:
```
limit((x^2-4)/(x-2),x=2)
```
Out[129]:
```
4
```

In [130]:
```
limit(f,x=0,dir="-")
```
Out[130]:
```
x |--> 2
```

In [131]:
```
f(x) = x*sin(1/x)
```

In [132]:
```
show(f.diff(x))
```

$$x \mapsto -\frac{\cos\left(\frac{1}{x}\right)}{x} + \sin\left(\frac{1}{x}\right)$$

In [133]:
```
diff(f(x),x,2) #1ST ORDER DERIVATIVE
```
Out[133]:
```
-sin(1/x)/x^3
```

In [134]:
```
f.diff(4) #or diff(f(x),x,4) gives 4th order derivative od f(x)
```
Out[134]:
```
x |--> -12*sin(1/x)/x^5 - 8*cos(1/x)/x^6 + sin(1/x)/x^7
```

problem :

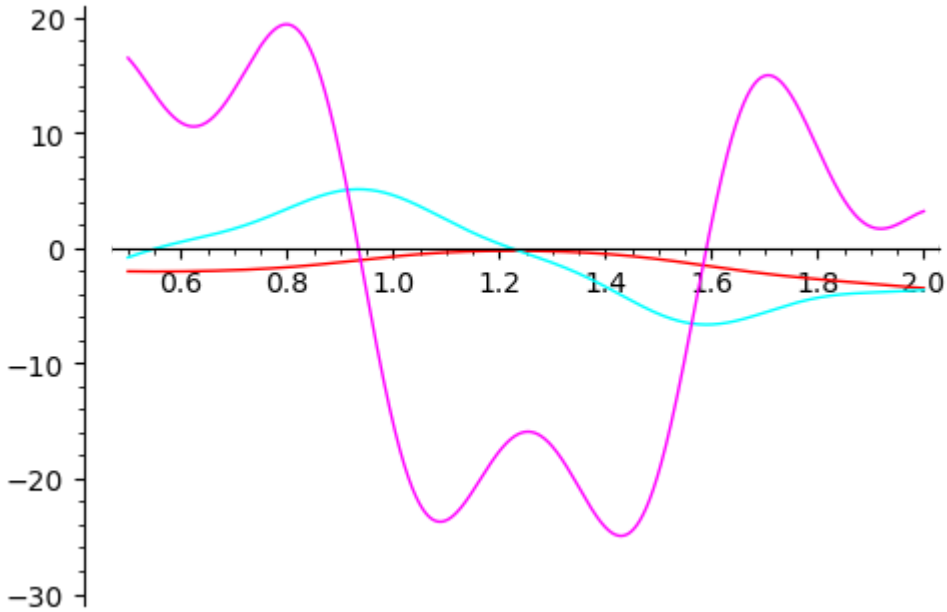
$Consideraf unctionf(x) = sin(cos(5x))) - e^{(}x - 1)^2$
$.plotthegraphoff(x)alongwithfirsttwoderivativein[0.5, 2].$

In [135]:
```
f(x) = sin(cos(5*x))-exp((x - 1)^2)
```

In [136]:
```
f1(x)=f.diff()
```

In [137]:
```
f2(x)=f.diff(2)
```

In [138]:
```
p = plot(f(x),0.5,2,color="red")
p1 = plot(f1(x),0.5,2,color="cyan")
p2 = plot(f2(x),0.5,2,color="magenta")
show(p+p1+p2, ymax=20, ymin=-30, figsize=5)
```



3.partial derivative

In [139]:
```
var('x,y')
f(x,y)=(x*y)/(x+y)
fx=f.diff(x) #Gives first oder derivative of f with respect to x
show(fx)
```

$$(x, y) \mapsto \frac{y}{x+y} - \frac{xy}{(x+y)^2}$$

In [140]:
```
fy=f.diff(y) #gives first order derivative of f with respect to y
show(fy)
```

$$(x, y) \mapsto \frac{x}{x+y} - \frac{xy}{(x+y)^2}$$

In [141]:
```
fxx=f.diff(x,2) #gives second order derivative of f with respect  to x
show(fxx)
```

$$(x, y) \mapsto -\frac{2\,y}{(x+y)^2} + \frac{2\,xy}{(x+y)^3}$$

In [142]:
```
fyy=f.diff(y,2)  #gives second order derivative of f with respect to y
show(fyy)
```

$$(x, y) \mapsto -\frac{2\,x}{(x+y)^2} + \frac{2\,xy}{(x+y)^3}$$

In [143]:
```
fxy=f.diff(x,y) # #gives derivative of f first wiith respect to x
show(fxy)
```

$$(x, y) \mapsto \frac{1}{x+y} - \frac{x}{(x+y)^2} - \frac{y}{(x+y)^2} + \frac{2\,xy}{(x+y)^3}$$

In [144]:
```
fyx=f.diff(y,x)   #gives derivative of f first wiith respect to y
show(fyx)
```

$$(x, y) \mapsto \frac{1}{x+y} - \frac{x}{(x+y)^2} - \frac{y}{(x+y)^2} + \frac{2\,xy}{(x+y)^3}$$

4.Implicit Derivative

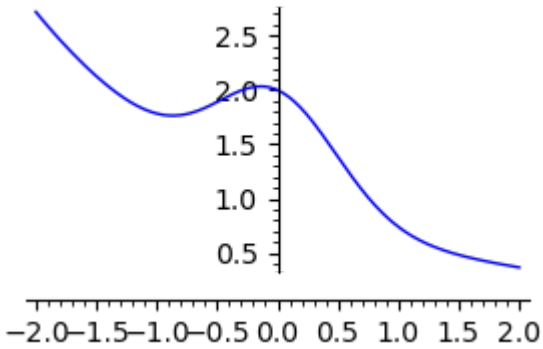
In [145]:
```
var('x,y')
f(x,y)=x^3 +y^3 - 6*x*y
y=function('y')(x)
temp=diff(f(x,y))
show(solve(temp,diff(y)))
```

$$\left[\frac{\partial}{\partial x} y(x) = -\frac{x^2 - 2\,y(x)}{y(x)^2 - 2\,x}\right]$$

Local Maximum and Minimum

In [146]:
```
f(x)= exp(-x/2) + exp(-2*x^2)
f.plot(-2,2,figsize =3)
```

Out[146]:



In [147]:
```
f.find_local_maximum(-1,1)
```

Out[147]: (2.0340673102820146, -0.13932332835389183)

In [148]:
```
f.find_local_minimum(-2,-0.5)
```

Out[148]: (1.7650250464715878, -0.866945722854257)

In [149]:
```
f.find_local_minimum(1,2)
```

Out[149]: (0.3682149119773104, 1.999999956179319)

In [150]:
```
f.find_local_maximum(1,2)
```

Out[150]:
```
(0.7418659187092465, 1.0000000286997572)
```

minima and maxima without constrains

$$x^3 - 3x + y^3 - 3y$$

In [151]:
```
var('x,y')
f(x,y)=x^3-3*x+y^3-3*y
fx=f.diff(x)
fy=f.diff(y)
R=f.diff(x,2)
T=f.diff(y,2)
S=f.diff(x,y)
H=R*T-S^2
```

In [152]:
```
show(fx)
show(fy)
```

$$(x, y) \mapsto 3\,x^2 - 3$$

$$(x, y) \mapsto 3\,y^2 - 3$$

In [153]:
```
show(R)
show(H)
```

$$(x, y) \mapsto 6\,x$$

$$(x, y) \mapsto 36\,xy$$

In [154]:
```
cp=solve([fx==0,fy==0],x,y)
show(cp)
```

$$[[x = 1, y = 1], [x = (-1), y = 1], [x = 1, y = (-1)], [x = (-1), y = (-1)]]$$

In [155]:
```
show(R.subs(cp[0]))
show(H.subs(cp[0]))
```

$$(x, y) \mapsto 6$$

$$(x, y) \mapsto 36$$

In [156]:
```
show(R.subs(cp[1]))
show(H.subs(cp[1]))
```

$$(x, y) \mapsto -6$$

$$(x, y) \mapsto -36$$

In [157]:
```
show(R.subs(cp[2]))
show(H.subs(cp[2]))
```

$$(x, y) \mapsto 6$$

$$(x, y) \mapsto -36$$

In [158]:
```
show(R.subs(cp[3]))
show(H.subs(cp[3]))
```

$$(x, y) \mapsto -6$$

$$(x, y) \mapsto 36$$

In [159]:
```
var('x,y')
f(x,y)=x^2+y^2
g(x,y)=x+y-1
L(x,y,t)=f(x,y)+t*g(x,y)
Lx=L.diff(x)
Ly=L.diff(y)
Lt=L.diff(t)
```

In [160]:
```
cp=solve([Lx==0,Ly==0,Lt==0],x,y,t)
show(cp)
```

$$\left[\left[x = \left(\frac{1}{2}\right), y = \left(\frac{1}{2}\right), t = (-1)\right]\right]$$

In [161]:
```
fext=f.subs(cp[0])
fext
```

Out[161]:
```
(x, y) |--> 1/2
```

Find the following limits

1.$limx \to 0(x-)22x1000$

1. $limx \to 0(x-)22x1000lim()x \to 0.5 - 2x - 1|2x - |3x2$

In [162]:
```
f(x) = x^2 - 2*x^1000
limit(f(x), x=0)
```

Out[162]:
```
0
```

In [163]:
```
f(x) = (2*x - 1) / (abs(2*x - 3) * x^2)
limit(f(x), x=0.5, dir='-')
```

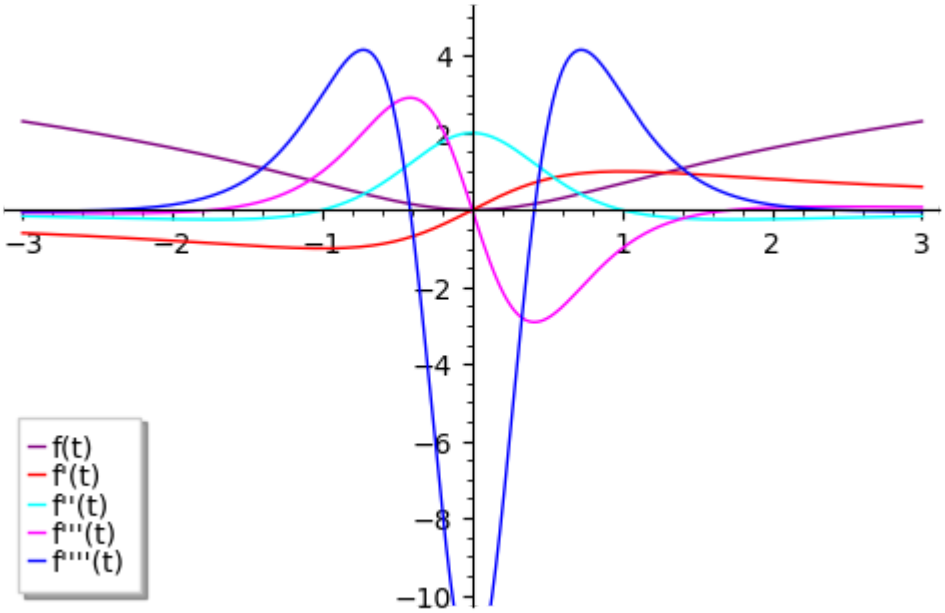
Out[163]:
```
0
```

Exercise 3.2 — Derivatives

Q1. $Firstfourderivativesoff(t) = ln(1 + t2)f(t) = \ln(1 + t^2)f(t) = ln(1 + t2),$
$plottedwithf(t)f(t)f(t)$

In [164]:
```
t = var('t')
f(t) = ln(1 + t^2)
f1(t) = f.diff(1)
f2(t) = f.diff(2)
f3(t) = f.diff(3)
f4(t) = f.diff(4)

p0 = plot(f(t), -3, 3, color='purple',   legend_label='f(t)')
p1 = plot(f1(t), -3, 3, color='red',   legend_label="f'(t)")
p2 = plot(f2(t), -3, 3, color='cyan', legend_label="f''(t)")
p3 = plot(f3(t), -3, 3, color='magenta',legend_label="f'''(t)")
p4 = plot(f4(t), -3, 3, color='blue', legend_label="f''''(t)")
show(p0+p1+p2+p3+p4, ymin=-10, ymax=5, figsize=5)
```



$Q2.First fourderivatives of f(x) = sin(3x) + ln(7x) + e - 2x f(x) = \sin(3x) + \ln(7x) + e^{-2x} f(x) = sin(3x) + ln(7x) + e^{-}2x$

In [165]:
```
f(x) = sin(3*x) + ln(7*x) + exp(-2*x)
show("f'(x)  =", f.diff(1))
show("f''(x) =", f.diff(2))
show("f'''(x)=", f.diff(3))
show("f''''(x)=",f.diff(4))
```

f'(x) $=x \mapsto \frac{1}{x} + 3\cos(3x) - 2e^{(-2x)}$

f''(x) $=x \mapsto -\frac{1}{x^2} + 4e^{(-2x)} - 9\sin(3x)$

f'''(x)$=x \mapsto \frac{2}{x^3} - 27\cos(3x) - 8e^{(-2x)}$

f''''(x)$=x \mapsto -\frac{6}{x^4} + 16e^{(-2x)} + 81\sin(3x)$

Exercise 3.3 — Partial Derivatives Q1.
$Let f(x,y) = ln(x2 + y2) f(x,y) = \ln(x^2 + y^2) f(x,y) = ln(x2 + y2). Showthat \partial 2f$
$\partial x2 + \partial 2f \partial y2 = 0 \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} = 0 \partial x2 \partial 2f + \partial y2 \partial 2f = 0$

In [166]:
```
var('x y')
f(x,y) = ln(x^2 + y^2)
fxx = f.diff(x, 2)
fyy = f.diff(y, 2)
result = simplify(fxx + fyy)
show("fxx + fyy =", result)
```

fxx + fyy = $(x,y) \mapsto -\frac{4\,x^2}{(x^2+y^2)^2} - \frac{4\,y^2}{(x^2+y^2)^2} + \frac{4}{x^2+y^2}$

$Q2. Findall first - orderpartialderivatives of w = xy2z3 + 43x - 7tan(4y)w$
$= xy^2z^3 + 43x - 7\tan(4y)w = xy2z3 + 43x - 7tan(4y)$

In [167]:
```
var('x y z')
w(x,y,z) = x*y^2*z^3 + 43*x - 7*tan(4*y)
show("dw/dx =", w.diff(x))
show("dw/dy =", w.diff(y))
show("dw/dz =", w.diff(z))
```

dw/dx = $(x,y,z) \mapsto y^2z^3 + 43$

dw/dy = $(x,y,z) \mapsto 2\,xyz^3 - 28\,\tan(4\,y)^2 - 28$

dw/dz = $(x,y,z) \mapsto 3\,xy^2z^2$

Exercise 3.4 — Implicit Derivatives

$Q1. Finddydx \frac{dy}{dx} dxdy\ for 2(x2 + y2)2 = 25(x2 - y2)$

In [168]:
```
x = var('x')
y = var('y')
f(x,y) = 2*(x^2 + y^2)^2 - 25*(x^2 - y^2)
y = function('y')(x)
temp = diff(f(x,y), x)
solve(temp, diff(y,x))
```

Out[168]:
```
[diff(y(x), x) == -(4*x^3 + 4*x*y(x)^2 - 25*x)/(4*y(x)^3 + (4*x^2 + 25)*y
(x))]
```

Q2. Find $dydx \frac{dy}{dx} dxdy\ fory = (2-x)2 \cdot x3y = (2-x)^2 \cdot x^3y = (2-x)2 \cdot x3$

In [169]:
```
x = var('x')
y = var('y')
f(x,y) = y - (2-x)^2 * x^3
y = function('y')(x)
temp = diff(f(x,y), x)
solve(temp, diff(y,x))
```

Out[169]:
```
[diff(y(x), x) == 5*x^4 - 16*x^3 + 12*x^2]
```

Exercise — Local Max & Min (Single Variable)

Q1.Find local max and min of $(x) = x4e - x$

In [170]:
```
f(x) = x^4 * exp(-x)
plot(f, -1, 6, figsize=4)

df  = f.diff()
d2f = f.diff(2)
cp  = solve(df == 0, x)
show("Critical points:", cp)

for pt in cp:
    val = d2f.subs(pt)
    show("At", pt, "-> f''=", val, "-> Max" if val < 0 else "Min" if val >
0 else "Inconclusive")

show("f(0) =", f(0))
show("f(4) =", f(4))
```

Critical points: $[x = 0, x = 4]$

At$x = 0$-> f’’=$x \mapsto 0$Inconclusive

At$x = 4$-> f’’=$x \mapsto -64\,e^{(-4)}$-> Max

f(0) =0

f(4) =$256\,e^{(-4)}$

Q2. Find speed v v that minimizes power $P = Av3 + BL2vP = Av^3 + \frac{BL^2}{v}P = Av3 + vBL2$

In [171]:
```
var('v A B L')
assume(A > 0, B > 0, L > 0, v > 0)
P(v) = A*v^3 + B*L^2/v
dP = P.diff(v)
v_min = solve(dP == 0, v)
show("Speed minimizing power:", v_min)
show("Minimum power:", P(v).subs(v_min[0]))
```

Speed minimizing power: $\left[v = i\left(\frac{1}{3}\right)^{\frac{1}{4}}\sqrt{L}\left(\frac{B}{A}\right)^{\frac{1}{4}}, v = -i\left(\frac{1}{3}\right)^{\frac{1}{4}}\sqrt{L}\left(\frac{B}{A}\right)^{\frac{1}{4}}, v = \left(\frac{1}{3}\right)^{\frac{1}{4}}\sqrt{L}\left(\frac{B}{A}\right)^{\frac{1}{4}}\right]$

Minimum power: $-i\left(\frac{1}{3}\right)^{\frac{3}{4}}AL^{\frac{3}{2}}\left(\frac{B}{A}\right)^{\frac{3}{4}} - \frac{3i\left(\frac{1}{3}\right)^{\frac{3}{4}}BL^{\frac{3}{2}}}{\left(\frac{B}{A}\right)^{\frac{1}{4}}}$

Exercise — Maxima & Minima Without Constraints

Q1. $f(x, y) = x^4 + y^4 - 4 * x^2 - 4 * y^2$

In [172]:
```
var('x y')
f(x,y) = x^4 + y^4 - 4*x^2 - 4*y^2
fx = f.diff(x);  fy = f.diff(y)
R = f.diff(x,2); S = f.diff(x,y); T = f.diff(y,2)
H = R*T - S^2
cp = solve([fx==0, fy==0], x, y)
show("Critical points:", cp)
for pt in cp:
    h_val = H.subs(pt)
    r_val = R.subs(pt)
    show(pt, "-> H=", h_val, ", fxx=", r_val)
```

$$\texttt{Critical points:}\ [[x=0, y=0], [x=-\sqrt{2}, y=0], [x=\sqrt{2}, y=0], [x=0, y=-\sqrt{2}], [x=0, y=\sqrt{2}], [x=-\sqrt{2}, y=-\sqrt{2}], [x=\sqrt{2}, y=-\sqrt{2}], [x=-\sqrt{2}, y=\sqrt{2}], [x=\sqrt{2}, y=\sqrt{2}]]$$

$[x=0, y=0]$ -> H= $(x, y) \mapsto 64$, fxx= $(x, y) \mapsto -8$

$[x=-\sqrt{2}, y=0]$ -> H= $(x, y) \mapsto -128$, fxx= $(x, y) \mapsto 16$

$[x=\sqrt{2}, y=0]$ -> H= $(x, y) \mapsto -128$, fxx= $(x, y) \mapsto 16$

$[x=0, y=-\sqrt{2}]$ -> H= $(x, y) \mapsto -128$, fxx= $(x, y) \mapsto -8$

$[x=0, y=\sqrt{2}]$ -> H= $(x, y) \mapsto -128$, fxx= $(x, y) \mapsto -8$

$[x=-\sqrt{2}, y=-\sqrt{2}]$ -> H= $(x, y) \mapsto 256$, fxx= $(x, y) \mapsto 16$

$[x=\sqrt{2}, y=-\sqrt{2}]$ -> H= $(x, y) \mapsto 256$, fxx= $(x, y) \mapsto 16$

$[x=-\sqrt{2}, y=\sqrt{2}]$ -> H= $(x, y) \mapsto 256$, fxx= $(x, y) \mapsto 16$

$[x=\sqrt{2}, y=\sqrt{2}]$ -> H= $(x, y) \mapsto 256$, fxx= $(x, y) \mapsto 16$

$Q2. f(x, y) = x^3 + y^3 - 63(x+y) + 12xy$

In [173]:
```
var('x y')
f(x,y) = x^3 + y^3 - 63*(x+y) + 12*x*y
fx = f.diff(x);  fy = f.diff(y)
R = f.diff(x,2); S = f.diff(x,y); T = f.diff(y,2)
H = R*T - S^2
cp = solve([fx==0, fy==0], x, y)
show("Critical points:", cp)
for pt in cp:
    h_val = H.subs(pt)
    r_val = R.subs(pt)
    show(pt, "-> H=", h_val, ", fxx=", r_val)
```

$$\texttt{Critical points:}\ [[x=3, y=3], [x=(-7), y=(-7)], [x=(-1), y=5], [x=5, y=(-1)]]$$

$[x=3, y=3]$ -> H= $(x, y) \mapsto 180$, fxx= $(x, y) \mapsto 18$

$[x=(-7), y=(-7)]$ -> H= $(x, y) \mapsto 1620$, fxx= $(x, y) \mapsto -42$

$[x=(-1), y=5]$ -> H= $(x, y) \mapsto -324$, fxx= $(x, y) \mapsto -6$

$[x=5, y=(-1)]$ -> H= $(x, y) \mapsto -324$, fxx= $(x, y) \mapsto 30$

Q3. $f(x,y) = x2 + y2 - 4x - 6y$

In [174]:
```
var('x y')
f(x,y) = x^2 + y^2 - 4*x - 6*y
fx = f.diff(x);  fy = f.diff(y)
R = f.diff(x,2); S = f.diff(x,y); T = f.diff(y,2)
H = R*T - S^2
cp = solve([fx==0, fy==0], x, y)
show("Critical points:", cp)
for pt in cp:
    show(pt, "-> H=", H.subs(pt), ", fxx=", R.subs(pt))
# H>0, fxx>0 → Minimum at (2,3), f(2,3) = -13
```

Critical points: $[[x = 2, y = 3]]$

$[x = 2, y = 3]$ -> H= $(x,y) \mapsto 4$, fxx= $(x,y) \mapsto 2$

Exercise — Lagrange Multipliers

Q1. Extremum of $f(x,y) = x^2 + y^2$ subject to $x + y = 5$

In [175]:
```
var('x y t')
f(x,y) = x^2 + y^2
g(x,y) = x + y - 5
L(x,y,t) = f(x,y) + t*g(x,y)
cp = solve([L.diff(x)==0, L.diff(y)==0, L.diff(t)==0], x, y, t)
show("Critical point:", cp)
show("Extremum value:", f.subs(cp[0]))
```

Critical point: $\left[\left[x = \left(\frac{5}{2}\right), y = \left(\frac{5}{2}\right), t = (-5)\right]\right]$

Extremum value: $(x,y) \mapsto \frac{25}{2}$

Q2. Extremum of $f(x,y) = xy$ subject to $x + y = 20$

In [176]:
```
var('x y t')
f(x,y) = x*y
g(x,y) = x + y - 20
L(x,y,t) = f(x,y) + t*g(x,y)
cp = solve([L.diff(x)==0, L.diff(y)==0, L.diff(t)==0], x, y, t)
show("Critical point:", cp)
show("Extremum value:", f.subs(cp[0]))
```

Critical point: $[[x = 10, y = 10, t = (-10)]]$

Extremum value: $(x,y) \mapsto 100$

Q3. Extremum of $f(x,y) = x + 2y$ subject to $x^2 + y^2 = 6$

In [177]:
```
var('x y t')
f(x,y) = x + 2*y
g(x,y) = x^2 + y^2 - 6
L(x,y,t) = f(x,y) + t*g(x,y)
cp = solve([L.diff(x)==0, L.diff(y)==0, L.diff(t)==0], x, y, t)
show("Critical points:", cp)
for pt in cp:
    show("f =", f.subs(pt))
```

$$\texttt{Critical points:}\ \left[\left[x = -\frac{1}{5}\sqrt{6}\sqrt{5}, y = -\frac{2}{5}\sqrt{6}\sqrt{5}, t = \frac{1}{12}\sqrt{6}\sqrt{5}\right], \left[x = \frac{1}{5}\sqrt{6}\sqrt{5}, y = \frac{2}{5}\sqrt{6}\sqrt{5}, t = -\frac{1}{12}\sqrt{6}\sqrt{5}\right]\right]$$

$$\texttt{f =} (x, y) \mapsto -\sqrt{6}\sqrt{5}$$

$$\texttt{f =} (x, y) \mapsto \sqrt{6}\sqrt{5}$$

What you learned?

In this experiment, I discovered how powerful SageMath can be for tackling calculus problems that we usually solve by hand. Instead of working through long manual steps, I was able to compute results quickly and even visualize them with graphs. I explored limits, including one-sided ones from the left and right, and calculated derivatives of complicated functions with ease. I moved on to partial derivatives, learning how to differentiate when functions depend on two variables like x and y. Implicit differentiation was another highlight, where y is hidden inside the equation but SageMath still handled it smoothly. I also found maximum and minimum values of functions without lengthy calculations, and for functions of two variables, I applied the Hessian method to classify points as maxima, minima, or saddle points. Finally, I learned about Lagrange Multipliers, which allow us to find extrema when a condition or constraint is attached. Altogether, SageMath turned what would normally be a time-consuming process into something faster, clearer, and more visual.